Lesson 1

Introduction to MS SQL Server

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Querying MS SQL Server

Create Database Objects (25%)

- Create and alter tables using T-SQL syntax
- Create and alter views
- Create and modify constraints on tables;
- Create and alter DML triggers.

Work with Data (35%)

- Query data by using SELECT statements.
- Implement sub-queries.
- Implement data types.
- Implement aggregate queries.

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Modify Data (20%)

- Create and alter Stored Procedures
- Modify data by using INSERT, UPDATE, and DELETE
- Combine datasets.(UNION ,MERGE)
- Work with functions.

Troubleshoot & Optimize (20%)

- Manage Transactions
- Mark a transaction; commit, and rollback;
- Implicit /explicit transactions;
- Isolation levels; scope and type of locks
- Use cursors; impact of scalar UDFs
- Combine multiple DML operations
- Implement error handling. Implement try/catch/throw;
- Optimize queries. understand statistics; read query plans;
- Plan guides; DMVs; hints; statistics IO;

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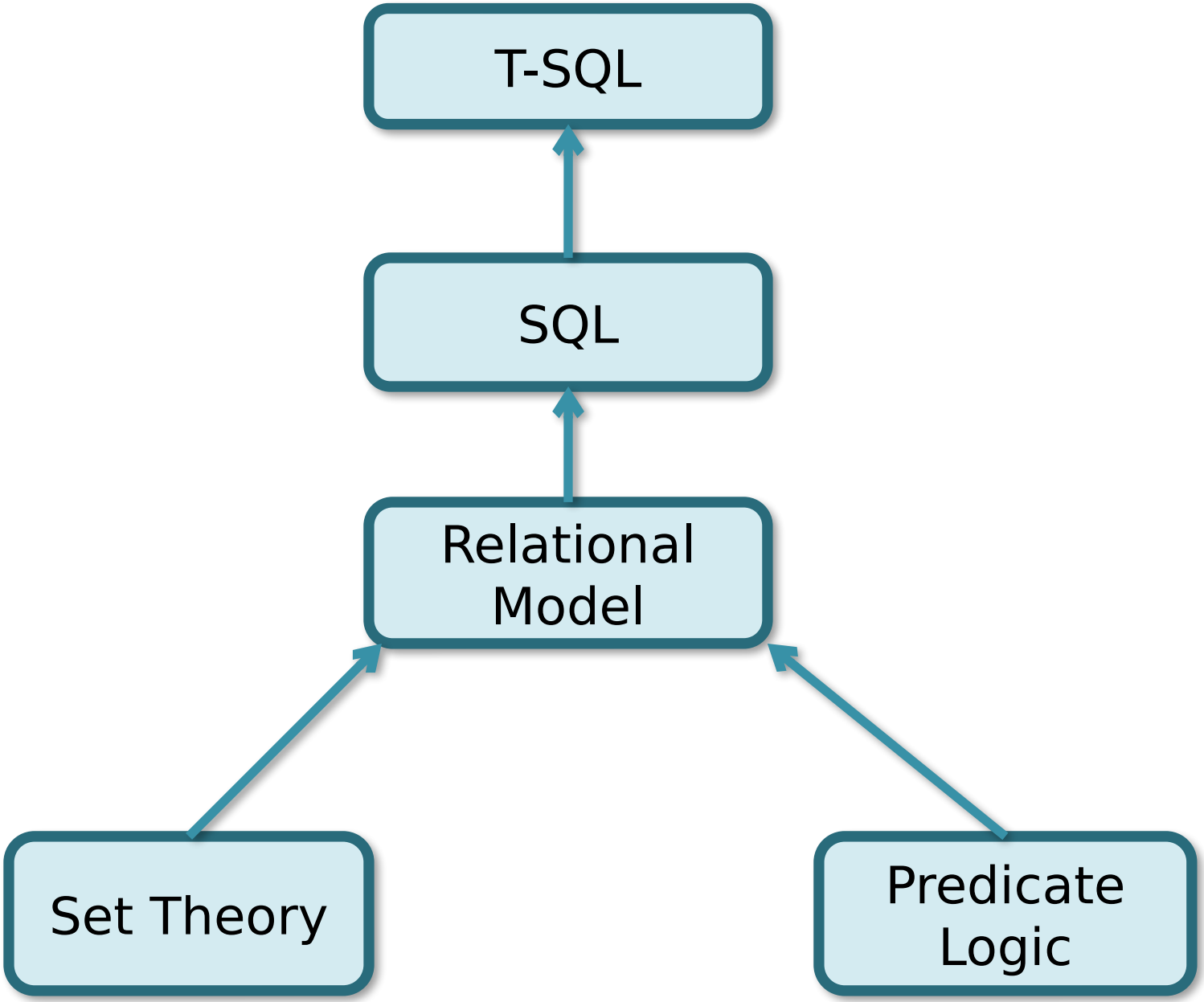

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Relational Algebra vs SQL

$$\pi_{e.name}\ (\ \sigma_{e.salary > m.salary}\ (\ \rho_e(employee) \bowtie_{e.manager = m.name} \rho_m(employee)\ )\ )$$

(a) Relational Algebra version

RANGE employee e;
RANGE employee m;
GET w (e.name): $\exists m((e.manager = m.name) \wedge (e.salary > m.salary))$

(b) Relational Calculus version

```
select e.name
from employee e, employee m
where e.manager = m.name and e.salary > m.salary
```

(c) Sequel (SQL) version

Figure 2. Three versions of the query, ''Find names of employees who earn more than their managers.''

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History of SQL Server

Microsoft and Sybase released version 1.0 in 1989.

However, the partnership between these two ended in the early 1990s.

Microsoft maintained ownership rights to the name SQL Server.

Since the 1990s, subsequent versions of SQL Server have been released including

SQL Server 2000, 2005, 2008, 2012, 2014, 2016, 2017, and 2019.

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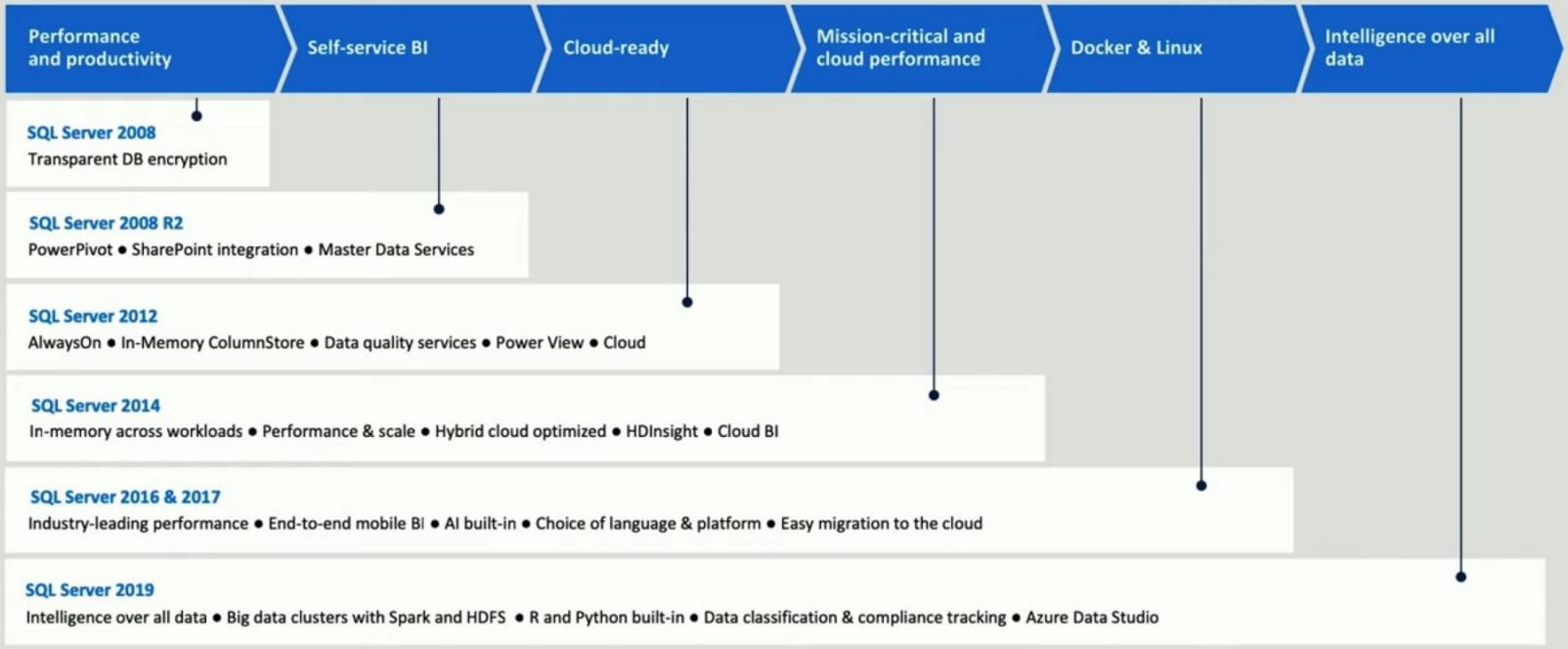
The evolution of SQL Server



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SQL Server Editions

- Enterprise edition
- Standard edition
- Web Edition
- Developer edition
- Express edition

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Enterprise Edition

- Maximum number of cores is subject to the OS being used
- Advanced high availability can be achieved with AlwaysOn
- Support for the new columnstore indexing feature
- Advanced auditing CDC
- Transparent Data Encryption (TDE)
- Compression and partitioning
- Includes all of the BI edition’s features and capabilities:
- Reporting
- Analytics
- Multidimensional BI semantic model
- Data-quality services
- Master data services
- In-memory tabular BI semantic model
- Self-service business intelligence

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Standard Edition

- Spatial support
- FileTable
- Policy-based management
- Corporate business intelligence
- Reporting
- Analytics
- Multidimensional BI semantic model
- Basic high availability can be achieved with AlwaysOn 2 Node Failover Clustering
- Up to 24 cores, up to 128 GB of RAM, one virtual machine, and two failover clustering nodes

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Developer Edition

- The Developer edition includes all of the features and functionality found in the Enterprise edition

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Web Edition

- Similar to Express edition
- Doesn’t have database size restrictions,
- Supports up to 4 CPU or 16 cores
- Supports up to 64 GB of memory

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Express Edition

- Limited to one processor or 4 cores
- 1 GB of memory,
- Maximum database size of 10 GB.
- Express is integrated with Microsoft Visual Studio

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SQL Server Instances

- Each instance of SQL Server contains its own set of :
  - Databases
  - Security credentials
  - Configuration settings
  - Windows services
  - All other SQL Server objects.
- Max 50 instances of SQL Server on a single machine, one default and the rest named instance

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- Default instance connection

  Uses the machine name to which the instance is installed.

- Named instance connection

  Uses the combination of the machine name and instance name, such as

<machinename>\
<instancename>

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- Reasons for more than one instance
  - Quality Assurance testing or Development
  - Need multiple service pack or patch levels
  - Different groups of administrators are allowed to access only a subset of databases within an organization
  - Multiple sets of SQL Server configuration options required

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SQL Server Configuration Manager

- SSCM is responsible for managing SQL Server services and protocols. i.e
  - Starting, stopping, pausing, and restarting a service
  - Changing service accounts and service account passwords
  - Managing the start-up mode of a service
  - Configuring service start-up parameters

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SQL Server Configuration

Console1 - [Console Root\SQL Server Configuration Manager (Local)\SQL Server Services]

File Action View Favorites Window Help

- Console Root
  - SQL Server Configuration Manager (Local)
    - SQL Server Services
    - SQL Server Network Configuration (32bit)
    - SQL Native Client 11.0 Configuration (32bit
      - Client Protocols
      - Aliases
    - SQL Server Network Configuration
      - Protocols for HUMBERBRIDGING
      - Protocols for SQL2K17
      - Protocols for SQL2K22
      - Protocols for TEST2K22
      - Protocols for SQL2K19
    - SQL Native Client 11.0 Configuration
      - Client Protocols
      - Aliases

Name	State	Start Mode	Log On As
SQL Server Analysis Services (SQL2K19)	Stopped	Manual	NT Service\MSOLAP$SQL2K19
SQL Server Analysis Services (SQL2K22)	Running	Automatic	NT Service\MSOLAP$SQL2K22
SQL Server (HUMBERBRIDGING)	Stopped	Manual	NT Service\MSSQL$HUMBERBF
SQL Server (SQL2K17)	Running	Automatic	NT Service\MSSQL$SQL2K17
SQL Server (SQL2K19)	Running	Manual	NT Service\MSSQL$SQL2K19
SQL Server (SQL2K22)	Running	Manual	NT Service\MSSQL$SQL2K22
SQL Server (TEST2K22)	Running	Automatic	NT Service\MSSQL$TEST2K22
SQL Full-text Filter Daemon Launcher (HUMBE...	Stopped	Manual	NT Service\MSSQLFDLauncher'
SQL Server Launchpad (SQL2K19)	Running	Automatic	NT Service\MSSQLLaunchpad$
SQL Server Launchpad (SQL2K22)	Running	Automatic	NT Service\MSSQLLaunchpad$
SQL Server Agent (HUMBERBRIDGING)	Stopped	Manual	NT Service\SQLAgent$HUMBEF
SQL Server Agent (SQL2K17)	Stopped	Manual	NT Service\SQLAgent$SQL2K17
SQL Server Agent (SQL2K19)	Stopped	Manual	NT Service\SQLAgent$SQL2K19
SQL Server Agent (SQL2K22)	Stopped	Manual	NT Service\SQLAgent$SQL2K22
SQL Server Agent (TEST2K22)	Stopped	Manual	NT Service\SQLAgent$TEST2K2
SQL Server Browser	Stopped	Manual	NT AUTHORITY\LOCALSERVICE

Actions

SQL Server Services

More Actions

Authentication Modes

Authentication mode for SQL Server can be set to either

- Windows Only (integrated security)
- Windows and SQL Server (mixed mode)

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SQL Server Management Studio

SSMS is an integrated environment for

- Accessing
- Configuring,
- Managing,
- Administering
- Developing

all components of SQL Server.

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Microsoft SQL Server Management

SQLQuery1.sql - DELL\SQL2K19.master (DELL\addyt (56))* - Microsoft SQL Server Management Studio

File Edit View Query Project Tools Window Help

New Query | master | Execute

Object Explorer

Connect

- DELL\SQL2K22 (SQL Server 16.0.1105.1 - DELL\addyt)
  - Databases
    - System Databases
    - Database Snapshots
    - AdventureWorks2019
    - DEMO
    - JOB_PORTAL_DB
    - ReportServer
    - ReportServerTempDB
  - Security
    - Logins
    - Server Roles
    - Credentials
    - Cryptographic Providers
    - Audits
    - Server Audit Specifications
  - Server Objects
  - Replication
  - Always On High Availability
  - Management
    - Policy Management
    - Data Collection
    - Resource Governor
    - Extended Events
    - Maintenance Plans
    - SQL Server Logs
    - Database Mail
    - Distributed Transaction Coordinator
  - Integration Services Catalogs
  - SQL Server Agent (Agent XPs disabled)
  - XEvent Profiler
- DELL\SQL2K19 (SQL Server 15.0.2000.5 - DELL\addyt)
  - Databases
  - Security
  - Server Objects
  - Replication
  - PolyBase
  - Always On High Availability
  - Management

SQLQuery1.sql - DE...r (DELL\addyt (56))*

```
select * from sysobjects
```

150 %

Results | Messages

	name	id	xtype	uid	info	status	base_schema_ver	replinfo	parent_ol
1	sp_MSalreadyhavegeneration	-1073624922	P	4	0	0	0	0	0
2	sp_MSwritemergeperfcounter	-1072815163	P	4	0	0	0	0	0
3	sp_drop_trusted_assembly	-1072667163	X	4	0	0	0	0	0
4	TABLE_PRIVILEGES	-1072372588	V	3	0	0	0	0	0
5	sp_replsetsyncstatus	-1071944761	X	4	0	0	0	0	0
6	sp_replshowcmds	-1070913306	P	4	0	0	0	0	0
7	sp_publishdb	-1070573756	P	4	0	0	0	0	0
8	fn_MSrepl_getsrvidfromdistdb	-1070030802	FN	4	0	0	0	0	0
9	sp_addqueued_artinfo	-1068897509	P	4	0	0	0	0	0
10	sp_replcounters	-1068756304	X	4	0	0	0	0	0
11	sp_MSget_subscription_dts_info	-1068452095	P	4	0	0	0	0	0
12	sp_help_spatial_geometry_index_xml	-1068265529	P	4	0	0	0	0	0
13	sp_password	-1067822458	P	4	0	0	0	0	0
14	dm_resource_governor_resource_pool_volumes	-1067705889	V	4	0	0	0	0	0

Query executed successfully. | DELL\SQL2K19 (15.0 RTM) | DELL\addyt (56) | master | 00:00:00 | 2,412 rows

Ready | Ln 1 | Col 14 | Ch 14 | INS

Transact-SQL

- Transact-SQL (TSQL) is the language used to create objects and access data in Microsoft SQL Server

- TSQL statement are divided into two distinct parts :
  - Data Manipulation Language
  - Data Definition Language

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- Data Manipulation Language (DML) is part of the TSQL language that allows insert, modify, and delete data in tables

- The core DML statements are INSERT, UPDATE, DELETE, and MERGE.

```
INSERT INTO Produce VALUES
(1,'Tomato','Vegetable'),
(2,'Pear',  'Fruit'),
(3,'Apple', 'Fruit');
```

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- Data Definition Language (DDL) is a part of TSQL that deals with creating database objects

- DDL statements are used to CREATE, ALTER, and DROP a SQL Server database object, such as a table, view, or stored procedure

```
CREATE TABLE Produce
(ProductID int PRIMARY KEY,
ProductName varchar (50),
ProductType varchar (20))
```

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- The SQL Server Management Studio (SSMS) user interface allows you to create an underlying DDL statement using the appropriate GUI component

- Not all functionality is available within the user interface!!!

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Most relational databases are *normalized*, with relationships defined between tables through *primary* and *foreign* keys

Customer		
CustomerID	FirstName	LastName
1	Dan	Drayton
2	Aisha	Witt
3	Rosie	Reeves

SalesOrderDetail			
OrderID	LineItemNo	ProductID	Quantity
1	1	3	1
2	1	2	5
2	2	3	1
3	1	1	1

SalesOrderHeader		
OrderID	OrderDate	CustomerID
1	1/1/2015	1
2	1/1/2015	3
3	1/2/2015	1

Product		
ProductID	Name	ListPrice
1	Widget	2.99
2	Gizmo	1.79
3	Thingybob	3.49

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Working with NULL values

- A NULL (value!) is the absence of data in a column.
- When the ANSI_NULLS setting is set to ON, the default configuration, you cannot equate values to NULL.
- If you try, the server returns “unknown”.

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```sql
SET ANSI_NULLS OFF;
SELECT DISTINCT region
FROM Production.Suppliers
WHERE region <> NULL;
```

```sql
SELECT DISTINCT region
FROM Production.Suppliers
WHERE region IS NOT NULL;
```

Same Result

```sql
SET ANSI_NULLS ON;
SELECT DISTINCT region
FROM Production.Suppliers
WHERE region <> NULL;
```

Unknown

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Logical Query processing order

- The logical query processing order is different from the order you write queries.

```
SELECT Country, HireDate
FROM HR.Employees
WHERE HireDate >= '20030101'
GROUP BY Country, HireDate
HAVING COUNT(*) > 1
ORDER BY country;
```

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Query Execution Order (Without UNION)

Clause	Result
FROM, JOIN, APPLY, and ON	The join is executed and the first query filter (the ON clause) is applied.
WHERE	The second query filter is applied.
GROUP BY and aggregate functions	Grouping and aggregation calculations are performed.
HAVING	The third query filter (filtering of the results of aggregate functions) is applied.
SELECT	Columns that should be returned by the query are selected.
ORDER BY	Results are sorted.
TOP	The fourth query filter is applied;
FOR XML	The tabular result returned by the SELECT statement is converted to XML

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Query Execution Order (With UNION)

Clause	Result
FROM, JOIN, APPLY, and ON	The join is executed and the first query filter (the ON clause) is applied.
WHERE	The second query filter is applied.
GROUP BY and aggregate functions	Grouping and aggregation calculations are performed.
TOP	The fourth query filter is applied;
UNION & SELECT	Columns that should be returned by the query are selected.
ORDER BY	Results are sorted.
FOR XML	The tabular result returned by the SELECT statement is converted to XML

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A query to retrieve all employees that were hired in or after the year 2003.

```
SELECT Country, YEAR(HireDate) AS
YearHired, COUNT(*) AS NumEmployees
FROM HR.Employees
WHERE HireDate >= '20030101'
GROUP BY Country, YEAR(HireDate)
HAVING COUNT(*) > 1
ORDER BY Country, YearHired DESC;
```

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- SQL Server first evaluates the from clause

	empid	lastname	firstname	title	titleofcourtesy	birthdate	hiredate	address	city	region	postalcode	country	phone	mgr
1	1	Davis	Sara	CEO	Ms.	1958-12-08 00:00:00.000	2002-05-01 00:00:00.000	7890 - 20th Ave. E., Apt. 2A	Seattle	WA	10003	USA	(206) 555-0101	NU
2	2	Funk	Don	Vice President, Sales	Dr.	1962-02-19 00:00:00.000	2002-08-14 00:00:00.000	9012 W. Capital Way	Tacoma	WA	10001	USA	(206) 555-0100	1
3	3	Lew	Judy	Sales Manager	Ms.	1973-08-30 00:00:00.000	2002-04-01 00:00:00.000	2345 Moss Bay Blvd.	Kirkland	WA	10007	USA	(206) 555-0103	2
4	4	Peled	Yael	Sales Representative	Mrs.	1947-09-19 00:00:00.000	2003-05-03 00:00:00.000	5678 Old Redmond Rd.	Redmond	WA	10009	USA	(206) 555-0104	3
5	5	Buck	Sven	Sales Manager	Mr.	1965-03-04 00:00:00.000	2003-10-17 00:00:00.000	8901 Garrett Hill	London	NULL	10004	UK	(71) 234-5678	2
6	6	Suurs	Paul	Sales Representative	Mr.	1973-07-02 00:00:00.000	2003-10-17 00:00:00.000	3456 Coventry House, Miner Rd.	London	NULL	10005	UK	(71) 345-6789	5
7	7	King	Russell	Sales Representative	Mr.	1970-05-29 00:00:00.000	2004-01-02 00:00:00.000	6789 Edgeham Hollow, Winchester Way	London	NULL	10002	UK	(71) 123-4567	5
8	8	Cameron	Maria	Sales Representative	Ms.	1968-01-09 00:00:00.000	2004-03-05 00:00:00.000	4567 - 11th Ave. N.E.	Seattle	WA	10006	USA	(206) 555-0102	3
9	9	Dolgopyatova	Zoya	Sales Representative	Ms.	1976-01-27 00:00:00.000	2004-11-15 00:00:00.000	1234 Houndstooth Rd.	London	NULL	10008	UK	(71) 456-7890	5

- Then the WHERE clause is evaluated

	empid	lastname	firstname	title	titleofcourtesy	birthdate	hiredate
1	4	Peled	Yael	Sales Representative	Mrs.	1947-09-19 00:00:00.000	2003-05-03 00:
2	5	Buck	Sven	Sales Manager	Mr.	1965-03-04 00:00:00.000	2003-10-17 00:
3	6	Suurs	Paul	Sales Representative	Mr.	1973-07-02 00:00:00.000	2003-10-17 00:
4	7	King	Russell	Sales Representative	Mr.	1970-05-29 00:00:00.000	2004-01-02 00:
5	8	Cameron	Maria	Sales Representative	Ms.	1968-01-09 00:00:00.000	2004-03-05 00:
6	9	Dolgopyatova	Zoya	Sales Representative	Ms.	1976-01-27 00:00:00.000	2004-11-15 00:

- This is why you can not refer to a column alias in WHERE clause

```
SELECT Country,
       YEAR(HireDate) AS YearHired
FROM HR.Employees
WHERE YearHired >= 2003;
```

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- Then the GROUP BY Clause is evaluated

```
SELECT Country, YEAR(HireDate) AS
YearHired, COUNT(*) AS NumEmployees
FROM HR.Employees
WHERE HireDate >= '20030101'
GROUP BY Country, YEAR(HireDate)
```

```
Group    Group                    Detail  Detail
         Detail
Country  YEAR(hiredate) empid     country
         hiredate
-------- -------------- ------    -------  ---------------
UK             2003          5     UK
2003-10-17
                       6     UK    2003-10-17
UK             2004          7     UK
2004-01-02
                             9     UK
2004-11-15
```

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- Filter Rows Based on the HAVING Clause

```
group          group          detail  detail  detail
country  YEAR(hiredate)       empid           country
         hiredate
--------  --------------  ------  -------  ---------------
UK             2003              5        UK      2003-10-17
                          6      UK       2003-10-17
UK             2004              7        UK      2004-01-02
                                 9        UK      2004-11-15
```

- Then SELECT Clause is evaluated

Results | Messages

	country	yearhired	numemployees
1	UK	2003	2
2	UK	2004	2

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- Finally handle presentation

```
SELECT Country,
YEAR(HireDate) AS YearHired,
COUNT(*) AS NumEmployees
FROM HR.Employees
WHERE HireDate >= '20030101'
GROUP BY Country, YEAR(HireDate)
HAVING COUNT(*) > 1
ORDER BY Country, YearHired DESC;
```

Results | Messages

	country	yearhired	numemployees
1	UK	2004	2
2	UK	2003	2

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